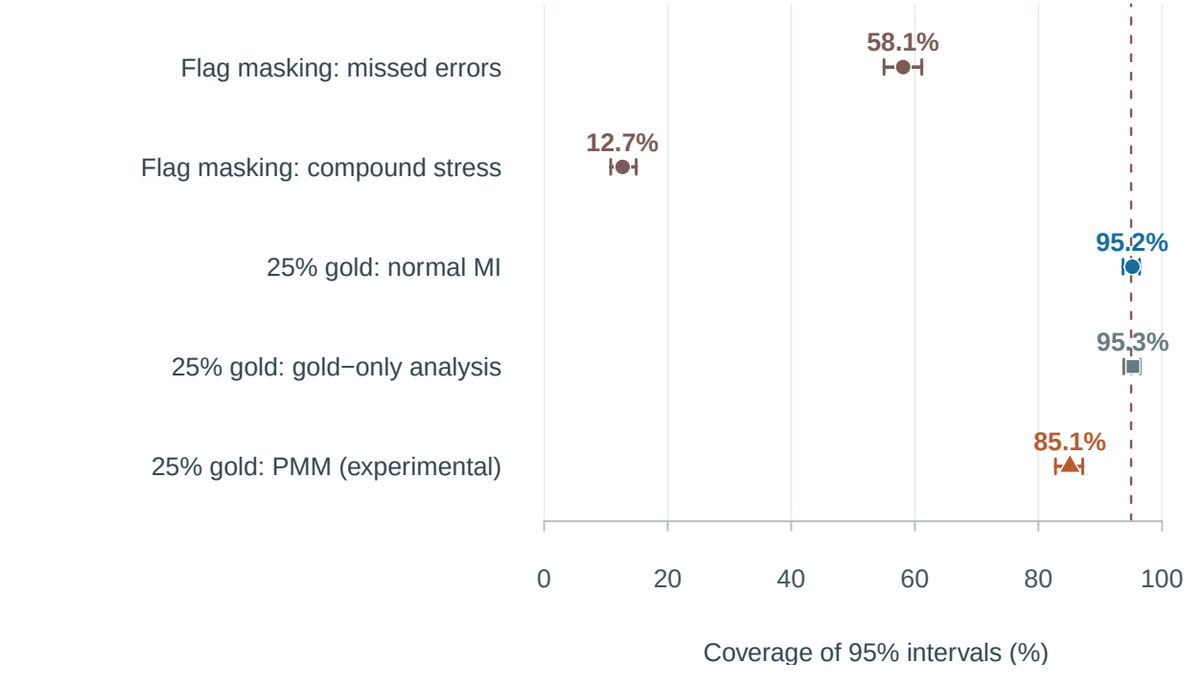
Measured references help; assumptions still matter

Held−out synthetic evidence | 20 imputations | Accurate reference assays provide additional information

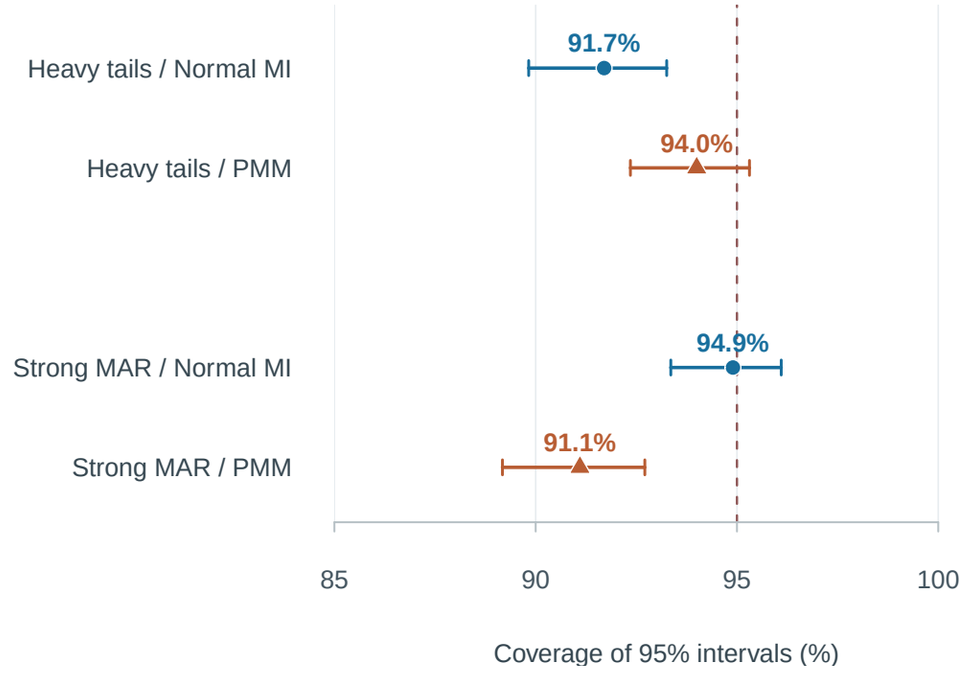
A New measurements change what is identifiable

Shared reference results shown once; 1,000 / 1,000 per row



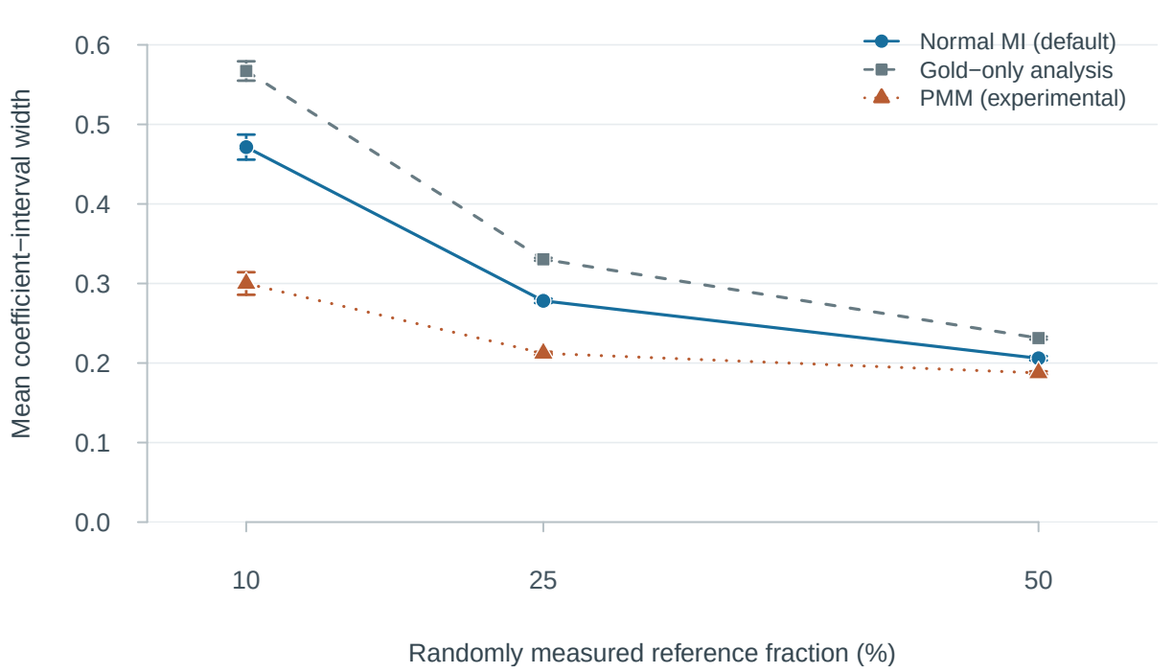
B Matching is not uniformly safer

Same predictors, no category; 1,000 / 1,000 per row



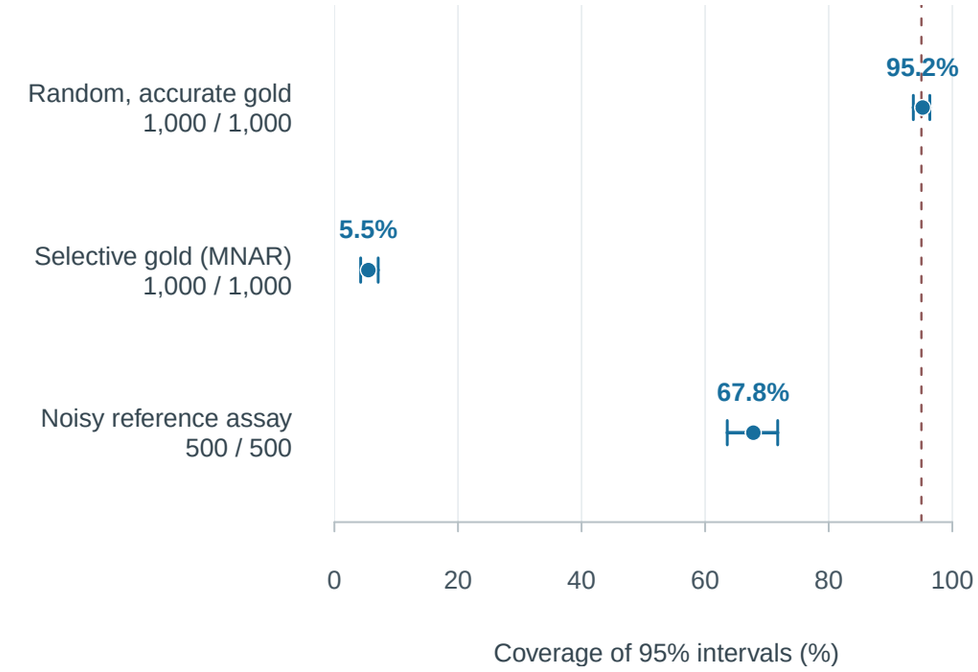
C More references buy precision

10% and 50%: 250 / 250; 25%: 1,000 / 1,000



D References must be credible

Normal MI; 25% expected reference sample



Coverage bars: 95% Wilson Monte Carlo intervals; dashed line: nominal 95%. Width bars: mean +/− 1.96 MCSE.

Counts are returned / requested. Panel B expands the coverage axis. Panel A reuses the same 1,000 reference cohorts.

Gold must be accurate and representative. Partial−reference PMM is experimental / sensitivity−only. All data are synthetic.